

PRAJWAL NARENDRA TIDKE

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Data engineer who has shipped production pipelines for a \$550B AUM institutional client (CPPIB) and built KALM, a clinical AI system spanning ingestion, RAG retrieval, and real-time crisis detection — from FastAPI backend to Vercel frontend. I work across the full data stack and am drawn to roles where robust data infrastructure and applied AI intersect. Currently pursuing an MS in Big Data Analytics at SDSU (4.0 GPA). Eligible for 36 months of US work authorization (STEM OPT) upon graduation.

EDUCATION

San Diego State University, MS, Big Data Analytics
(GPA: 4.0 / 4.0)

08/2024 – 05/2026 | San Diego, CA

EXPERIENCE

San Diego State University, Research Assistant

02/2025 – Present | San Diego, CA

- Built KALM, a clinical mental-health chatbot (FastAPI + Azure OpenAI GPT-4.1 + ChromaDB) ingesting DSM-5 and workplace safety docs (SAMHSA, OSHA, NIOSH) via metadata-tagged chunking and topic-filtered vector retrieval; deployed on Render with a Vercel frontend.
- Engineered a real-time crisis-signal interceptor and ALGEE-stage state machine across 4 AI personas. The system handles live crisis escalation, so security was non-negotiable: AES-128 encryption, JWT/Supabase auth, prompt injection guards, and GDPR-compliant deletion endpoints.
- Evaluated KBLaM for clinical LLM augmentation — built 1,000+ KB triples and QA pairs; pivoted to RAG after evaluation showed insufficient accuracy gains (Microsoft later discontinued the approach).
- Built a WGAN biosignal pipeline (PyTorch + Hugging Face) generating 52K synthetic training samples; holdout evaluation showed 14.7% MAE reduction over real-data-only baseline (5-fold cross-validation, logged in W&B).

TresVista Analytics, Associate Data Engineer

10/2023 – 08/2024 | Pune, India

- CPPIB's portfolio reporting ran on a Databricks + Airflow stack managing \$550B+ AUM; I architected and managed the ETL/ELT workflow layer and dimensional schema design, with outputs surfaced in Power BI & Valsys dashboards used daily by portfolio managers.
- Redesigned SQS→S3 fan-out logic and PySpark partition strategy for a 1.2 TB/day ingestion layer; Databricks job-run history showed 38% median latency drop and 19% throughput gain over a 6-week post-deploy window (team effort — owned the partition redesign and SQS batching layer).
- Applied Z-ordering on 4 high-cardinality Delta tables and rewrote 11 partition-unaware queries; Databricks SQL query history showed median execution time fell from 4.2 min to 2.7 min across all Power BI & Valsys dashboards (36% reduction, measured over 30-day pre/post window).
- Deployed Great Expectations with pytest CI/CD gates; 91% expectation coverage across 3 workstreams (measured as % of schema columns and row-count assertions passing in Great Expectations data docs) — Maintained 100% SLA compliance and zero schema violations across 6 months post-launch.
- Built a VBA plugin integrating S&P Global and Visible Alpha feeds directly into Excel financial models, automated workflows, eliminating manual data updates for 3 analysts across 2 model types; team's estimate of ~22 hours/week saved, based on pre/post time-tracking over a 4-week period.

LTIMindtree, Senior Data Engineer (enterprise delivery · promoted L3, 2021)

07/2020 – 10/2023 | Pune, India

- Five enterprise client engagements over 3 years — BFSI and manufacturing verticals — in a 3-person Agile team. Covered the full delivery lifecycle: sprint planning, code reviews, governance audits, and QBRs with client stakeholders.
- Optimized SAP BODS pipelines across SAP ECC, HANA, SQL Server, and Oracle via predicate pushdown and composite index tuning; end-of-month reporting cycles for 2 enterprise clients dropped from 18-24 hrs to under 3 hrs (measured via Tidal job monitor pre/post across 3 release cycles).
- Owned Python automation layer for 12 Oracle→Snowflake schema translations (AWS Glue); my workstream eliminated 3 manual handoffs and reduced transform runtime by 41% vs. baseline BODS jobs. Overall migration reduced client infrastructure and licensing spend by ~48% (figure sourced from client finance team post-migration audit).
- Automated 7 recurring ELT workflows in Python (previously manual, run by 2 ops staff); time-tracking logs showed 58 hrs/month of manual effort eliminated, equivalent to 0.9 FTE annually — validated across 3 consecutive monthly ops reviews.

SKILLS

Data Engineering & Cloud: PySpark (1.2 TB+/day production), Apache Airflow (DAG design + CI/CD gating), Databricks (Delta Lake, SQL Tuning, job orchestration), Snowflake (schema migration, query optimization), AWS (S3, SQS, SNS, Glue), PostgreSQL, Oracle, SAP BODS/HANA

AI & Machine Learning: RAG Architectures, LangChain, ChromaDB, Azure OpenAI API, PyTorch, Hugging Face, LLM Augmentation, Vector Retrieval, Spatial Clustering (DBSCAN)

Full-Stack & DevOps: FastAPI, Next.js, React, Sanity CMS, Prisma ORM, Docker, Git, CI/CD (GitHub Actions/pytest), Great Expectations

Languages: Python (Advanced), SQL (Advanced), TypeScript, JavaScript, VBA

Architectural Patterns: Medallion Architecture (Bronze/Silver/Gold), ETL/ELT Pipeline Design, Dimensional Modeling, Data Lakehouse, Real-time Crisis Detection, Data Governance

Visualization & BI: Power BI, Tableau, Looker, SAP BI (report consumption, LTIMindtree), Valsys

PROJECTS

AI-Powered Tax Advisory Platform 

12/2025 – Present

- Architected a full-stack AI application using Next.js, React, TypeScript, and OpenAI Assistants API with thread-based persistent memory in PostgreSQL and streaming inference via Vercel AI SDK — delivering cited legal guidance with sub-second response times.
- Engineered freemium infrastructure with NextAuth.js OAuth, RBAC rate limiting via Prisma ORM, and Sanity headless CMS with Incremental Static Regeneration — enabling zero-downtime content updates.

TrafficSensAI — Predictive Accident Analysis 

08/2024 – 12/2024

- Built an end-to-end ML pipeline using DBSCAN spatial clustering and Random Forest classification on 487K US incident records (2016–2023, Kaggle source); holdout test set (80/20 split) showed 86.3% weighted F1 for accident severity classification — deployed in Docker with an interactive hotspot map.